

Tutorial-3 (Analysis of LTI system in freq. domain)

1. (a) Magnitude and phase response.

$$e^{j\omega} = \cos\omega + j\sin\omega$$

eulers.

Solution:

We have, $H(z) = \frac{1-z^{-1}}{1-0.5z^{-1}}$

put $z = e^{j\omega}$

$$H(e^{j\omega}) = \frac{1 - e^{-j\omega}}{1 - 0.5e^{-j\omega}}$$

$$= \frac{1 - \cos\omega + j\sin\omega}{1 - 0.5(\cos\omega + j\sin\omega)}$$

$$= \frac{(1 - \cos\omega) + j\sin\omega}{(1 - 0.5\cos\omega) + (0.5\sin\omega)j}$$

For magnitude Response

So,

$$|H(e^{j\omega})| = \frac{\sqrt{(1 - \cos\omega)^2 + (\sin\omega)^2}}{\sqrt{(1 - 0.5\cos\omega)^2 + (0.5\sin\omega)^2}}$$

$$= \frac{\sqrt{1 - 2\cos\omega + \cos^2\omega + \sin^2\omega}}{\sqrt{1 - \cos\omega + 0.25\cos^2\omega + 0.25\sin^2\omega}}$$

$$= \frac{\sqrt{2 - 2\cos\omega}}{\sqrt{1.25 - \cos\omega}}$$

This is the required magnitude response of $H(e^{j\omega})$ or $H(z)$

For phase Response

$$\angle H(e^{j\omega}) = \angle (1 - e^{-j\omega}) - \angle (1 - 0.5e^{-j\omega})$$

$$= \angle (\cos\omega + j\sin\omega) - \angle (1 + 0.5\cos\omega - 0.5j\sin\omega)$$

$$= -0.5\cos\omega + 0.5j\sin\omega$$

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{-0.5\cos\omega}{0.5\sin\omega} \right)$$

$$\text{phase: } \angle H(e^{j\omega}) = -\tan^{-1} \left(\frac{0.5 \sin \omega}{-0.5 \cos \omega} \right)$$

$$= -\tan^{-1} (-\tan \omega)$$

$$= \omega$$

for plot: Magnitude response

ω	0	$\pi/3$	$\pi/2$	$3\pi/4$	$5\pi/6$	π
$H(\omega)$	0	1.33	1.30	1.33	1.04	1.31

for phase response:

ω	0
$\angle H(e^{j\omega})$	0

$$\textcircled{b} \quad H(z) = \frac{1 - 2z^{-1}}{1 - 0.6z^{-1} + 0.2z^{-2}}$$

put $z = e^{j\omega}$

$$H(e^{j\omega}) = \frac{1 - 2e^{-j\omega}}{1 - 0.6e^{-j\omega} + 0.2e^{-2j\omega}}$$

using Euler's, $e^{-j\omega} = \cos\omega - j\sin\omega$

$$H(e^{j\omega}) = \frac{1 - 2(\cos\omega - j\sin\omega)}{1 - 0.6(\cos\omega - j\sin\omega) + 0.2(\cos 2\omega - j\sin 2\omega)}$$

$$= \frac{1 - 2\cos\omega + 2j\sin\omega}{1 - 0.6\cos\omega + j(0.6\sin\omega) + 0.2\cos 2\omega - j(0.2\sin 2\omega)}$$

$$= \frac{(1 - 2\cos\omega + 2j\sin\omega)}{1 + (0.6\cos\omega + 0.2\cos 2\omega) + j(0.6\sin\omega - 0.2\sin 2\omega)}$$

magnitude of numerator:

$$\begin{aligned} &\Rightarrow \sqrt{(1 - 2\cos\omega)^2 + (2\sin\omega)^2} \\ &= \sqrt{1 - 4\cos\omega + 4\cos^2\omega + 4\sin^2\omega} \\ &= \sqrt{5 - 4\cos\omega} \end{aligned}$$

denominator:

$$\Rightarrow \sqrt{(1 - 0.6\cos\omega + 0.2\cos 2\omega)^2 + (0.6\sin\omega - 0.2\sin 2\omega)^2}$$

$$\therefore |H(\omega)| = \frac{\sqrt{5 - 4\cos\omega}}{\sqrt{(1 - 0.6\cos\omega + 0.2\cos 2\omega)^2 + (0.6\sin\omega - 0.2\sin 2\omega)^2}}$$

$$\angle H(\omega) = \tan^{-1} \left(\frac{\text{img}}{\text{Real}} \right) = \tan^{-1} \left(\frac{1 - 2e^{-j\omega} - 1.4 \cdot 0.6e^{-j\omega} - 0.2e^{-2j\omega}}{-1.4e^{-j\omega} - 0.2e^{-2j\omega}} \right)$$

$$= \tan^{-1} \left(\frac{-1.4(\cos\omega - j\sin\omega) - 0.2(\cos 2\omega - j\sin 2\omega)}{-1.4\cos\omega + j(1.4\sin\omega) - 0.2\cos 2\omega + j(0.2\sin 2\omega)} \right)$$

$$= \tan^{-1} \left(\frac{(-1.4\cos\omega - 0.2\cos 2\omega) + j(1.4\sin\omega + 0.2\sin 2\omega)}{-1.4\cos\omega - 0.2\cos 2\omega} \right)$$

$$\angle H(\omega) = \tan^{-1} \left(\frac{1.4\sin\omega + 0.2\sin 2\omega}{-1.4\cos\omega - 0.2\cos 2\omega} \right)$$

ω	0	$\pi/5$	$2\pi/5$	$3\pi/5$	$4\pi/5$	π
$H(\omega)$						
$\phi(\omega)$						

Q. Output sequence :

$$h[n] = 0.5^n u[n] \quad \text{--- (1)}$$

$$x[n] = A e^{j\pi n/2}, \quad -\infty < n < \infty \quad \text{--- (2)}$$

Taking Z transform of (1)

$$H(z) = Z(0.5^n u[n])$$

$$= \frac{1}{1 - 0.5z^{-1}}, \quad \text{Roc: } |z| > 0.5$$

$$X(z) = Z(A e^{j\pi n/2})$$

Let $r = e^{j\pi/2}$

$$X(z) = Z(A r^n)$$

$$= A Z(r^n)$$

$$= A \frac{1}{1 - r}$$

$$= \frac{A}{1 - e^{j\pi/2}}$$

$$\therefore Y(z) = \frac{X(z)}{H(z)} = \frac{A}{1 - e^{j\pi/2}} \times (1 - 0.5z^{-1})$$

$$Y(z) = A \frac{1 - 0.5z^{-1}}{1 - e^{j\pi/2}}$$

Inverse Z transform of $Y(z)$:

$$z^{-1}(Y(z)) = z^{-1} \left(A \frac{1 - 0.5z^{-1}}{1 - e^{j\pi/2}} \right)$$

$\equiv$

z^{-n} delay



Chap 4 :

1. Direct form - I :

$$y[n] = -0.9y[n-1] + 0.6y[n-2] - 0.1y[n-3] + 3x[n] + 2x[n-3] \quad \text{--- (1)}$$

For transfer function $H(z)$:

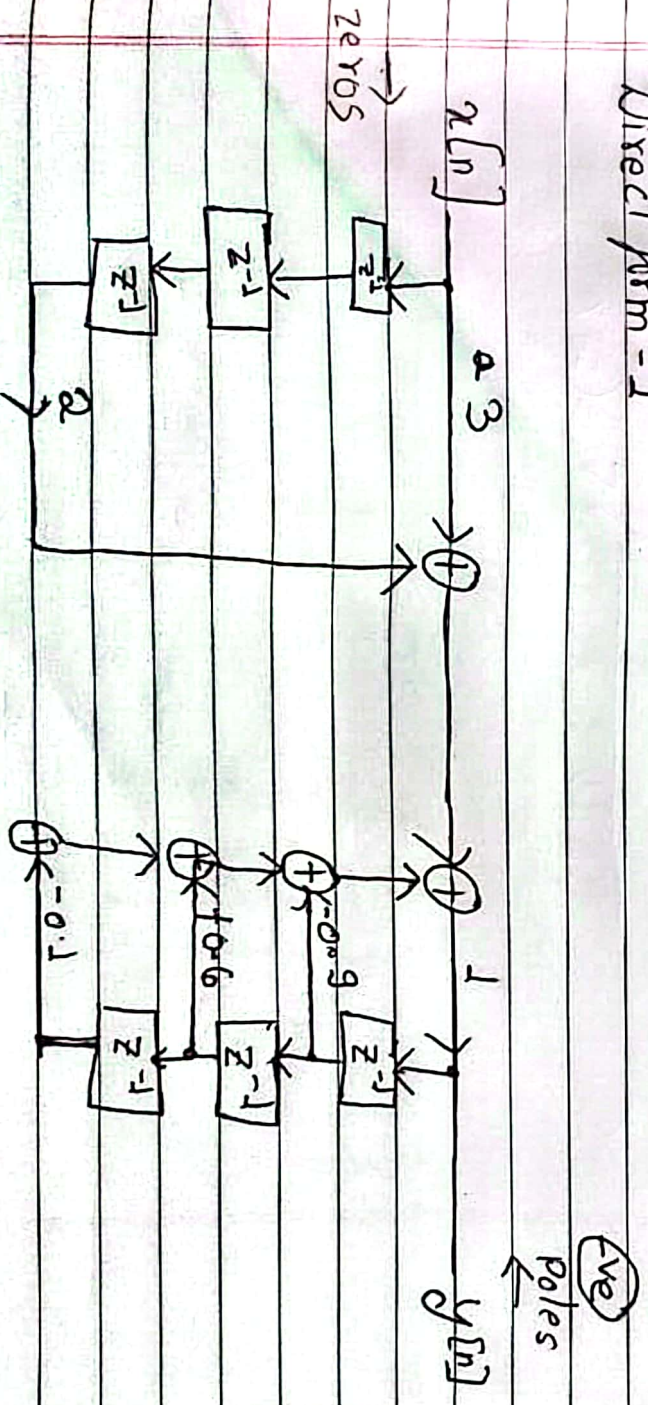
Take Z transform of (1)

$$Y[z] = -0.9Y[z]z^{-1} + 0.6Y[z]z^{-2} - 0.1Y[z]z^{-3} + 3X[z] + 2X[z]z^{-3}$$

$$Y[z](1 + 0.9z^{-1} - 0.6z^{-2} + 0.1z^{-3}) = X[z](3 + 2z^{-3})$$

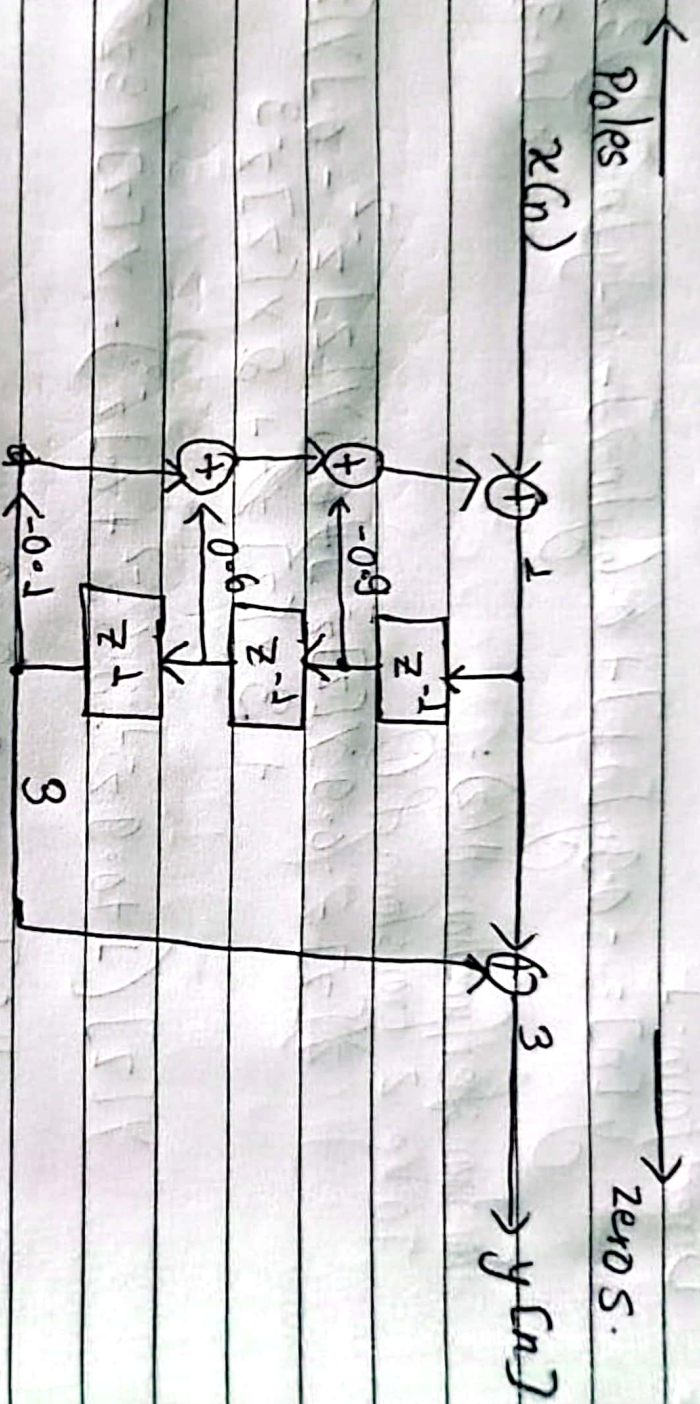
$$H[z] = \frac{3 + 2z^{-3}}{1 + 0.9z^{-1} - 0.6z^{-2} + 0.1z^{-3}}$$

Direct form - I



Direct form I Structure.

Direct form II Structure



2. CASCADE Structure

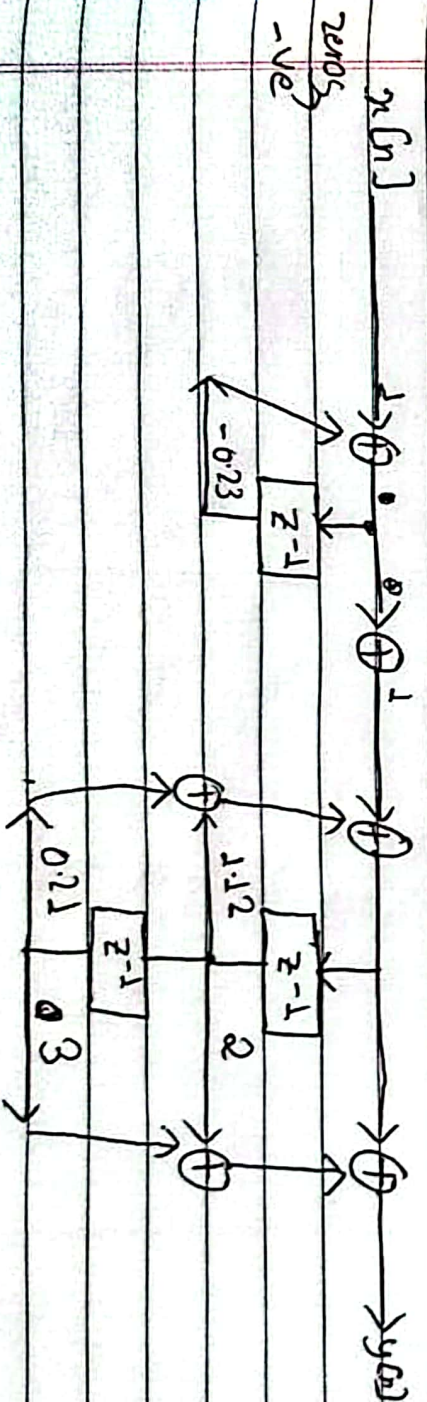
$$H(z) = \frac{1 + 2z^{-1} + 3z^{-2}}{1 - 0.9z^{-1} + 0.6z^{-2} + 0.2z^{-3}}$$

$$= \frac{1 + 2z^{-1} + 3z^{-2}}{(1 + 0.23z^{-1})(1 - (0.56 - 0.73j)z^{-1})(1 - (0.56 + 0.73j)z^{-1})}$$

$$= \frac{1 + 2z^{-1} + 3z^{-2}}{(1 + 0.23z^{-1})(1 - (0.56 + 0.73j)z^{-1})(1 - (0.56 - 0.73j)z^{-1})}$$

$$= \frac{1 + 2z^{-1} + 3z^{-2}}{(1 + 0.23z^{-1})(1 - 1.08z^{-1} + (0.56)^2 z^{-2} - (0.73j)^2 z^{-2})}$$

$$\begin{aligned} &= \frac{1}{(1 + 0.23z^{-1})(1 - 1.12z^{-1} + 0.21z^{-2})} \\ &= H_1 \times H_2 \end{aligned}$$



Lattice Structure:-

$$\textcircled{a} \quad \frac{1 + 2z^{-1} + 3z^{-2}}{1 - 0.9z^{-1} + 0.6z^{-2}}$$

Here, denominator is second order so $m=2$
 ~~$k_2 = q_2(0) = 1$~~ $k_2 = q_2(2) = 0.6$

For lattice part:-

$$A_2(z) = 1 - 0.9z^{-1} + 0.6z^{-2}$$

$$A_1(z) = A_2(z) - k_2 B_2(z)$$
$$1 - k_2^2$$

$$= \frac{1 - 0.9z^{-1} + 0.6z^{-2} - (0.6)(0.6 - 0.9z^{-1} + z^{-2})}{1 - 0.6^2}$$

$$= \frac{1 - 0.9z^{-1} + 0.6z^{-2} - 0.36 + 0.54z^{-1} - 0.6z^{-2}}{1 - 0.36}$$

$$= \frac{(1 - 0.36) - 0.36z^{-1}}{0.64}$$

$$= \frac{0.64 - 0.36z^{-1}}{0.64}$$

$$= 1 - 0.56z^{-1}$$

$$k_1 = -0.56$$

for ladder part:

$$C_Q(z) = 1 + 2z^{-1} + 3z^{-2}$$

$$V_Q = 4.3$$

$$C_1(z) = C_Q(z) - V_Q B_Q(z)$$

$$= 1 + 2z^{-1} + 3z^{-2} - 3(0.6 - 0.9z^{-1} + z^{-2})$$

$$= 1 + 2z^{-1} + 3z^{-2} - 1.8 + 2.7z^{-1} - 3z^{-2}$$

$$= -0.8 + 4.7z^{-1}$$

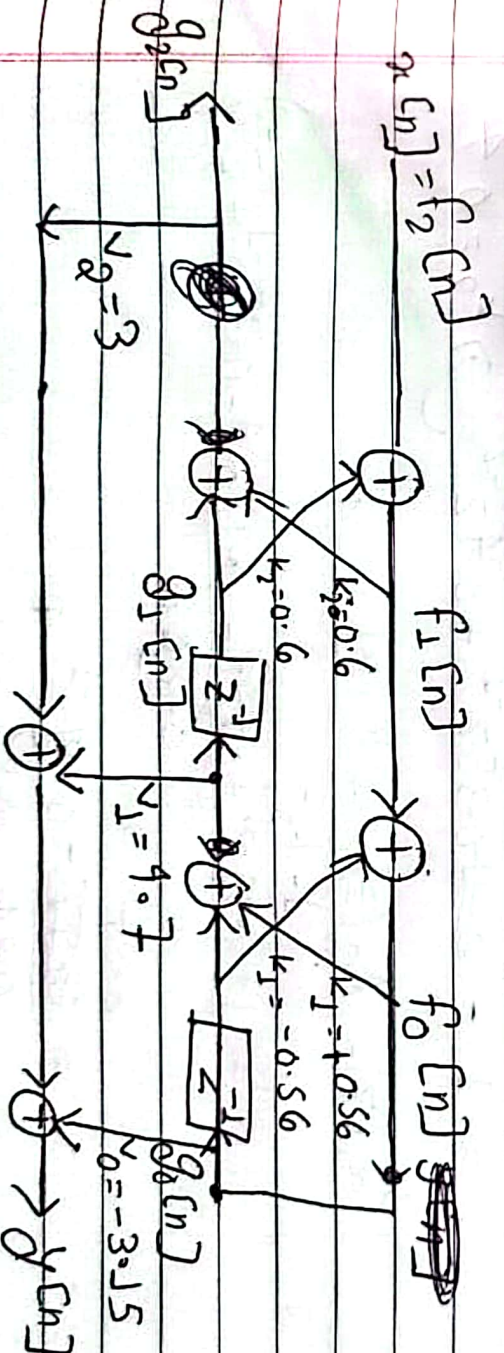
$$V_1 = 4.7$$

$$C_0(z) = C_1(z) - V_1 B_1(z)$$

$$\text{lattice ladder.}$$

$$= -0.8 + 4.7z^{-1} - (4.7)(0.5z^{-1})$$

$$= -3.15$$



lattice ladder

$$(b) \quad H(z) = \frac{1 + 2z^{-1} + 3z^{-2} + z^{-3}}{1 - 0.9z^{-1} + 0.6z^{-2} + 0.2z^{-3}}$$

Here, $A_3(z) = 1 - 0.9z^{-1} + 0.6z^{-2} + 0.2z^{-3}$
 $k_3 = 0.2$

$$A_2(z) = \frac{A_3(z) - k_3 A_2(z)}{1 - k_3^2}$$

$$= \frac{1 - 0.9z^{-1} + 0.6z^{-2} + 0.2z^{-3} - 0.2(0.2 + 0.6z^{-1} - 0.9z^{-2} + z^{-3})}{1 - 0.4}$$

$$= \frac{(1 - 0.4) - 0.9z^{-1} - 0.12z^{-1} + 0.6z^{-2} + 0.18z^{-2}}{0.6}$$

$$= 1 - 0.35z^{-1} + 0.4z^{-2} \rightarrow k_2 = 0.4$$

$$A_1(z) = \frac{A_2(z) - k_2 A_1(z)}{1 - k_2^2}$$

$$= \frac{1 - 0.35z^{-1} + 0.4z^{-2} - 0.4(0.4 - 0.35z^{-1} + z^{-2})}{1 - 0.16}$$

$$= \frac{0.84 - 0.28z^{-1} + 0.24z^{-2} - 0.16z^{-2}}{0.84}$$

$$= 1 - 0.25z^{-1}$$

$$k_1 = -0.25$$

for ladder port

$$C_3(z) = 1 + 2z^{-1} + 3z^{-2} + z^{-3}$$

$$v_3 = 1$$

$$\begin{aligned} C_2(z) &= C_3(z) - v_3 B_3(z) \\ &= 1 + 2z^{-1} + 3z^{-2} + z^{-3} - 1 - 3z^{-1} - 2z^{-2} - z^{-3} \\ &= -z^{-1} + z^{-2} \end{aligned}$$

$$\begin{aligned} C_1(z) &= C_2(z) - v_2 B_2(z) \\ &= 1 + 2z^{-1} + 3z^{-2} + z^{-3} - 1(0.2 + 0.6z^{-1} + 0.9z^{-2} + z^{-3}) \\ &= (1 - 0.2) + (2 - 0.6)z^{-1} + (3 - 0.9)z^{-2} \\ &= 0.8 + 1.4z^{-1} + 3.9z^{-2} \end{aligned}$$

$\hookrightarrow v_2 = 3.9$

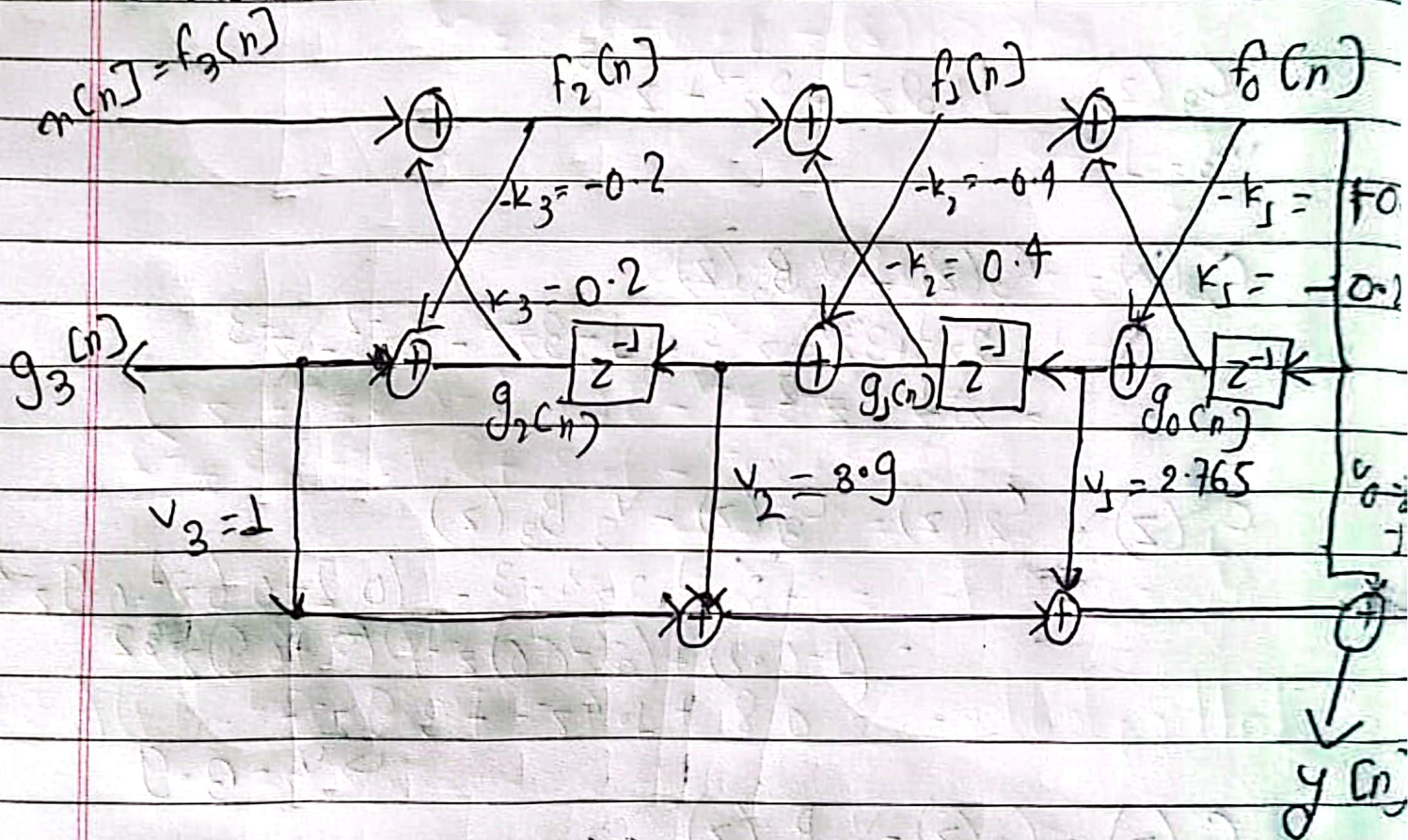
$$\begin{aligned} C_1(z) &= C_2(z) - v_2 B_2(z) \\ &= 0.8 + 1.4z^{-1} + 3.9z^{-2} - 3.9(0.4 - 0.35z^{-1} + z^{-2}) \\ &= 0.8 + 1.4z^{-1} + 3.9z^{-2} - 1.56 + 1.365z^{-1} - 3.9z^{-2} \\ &= -0.76 + 2.765z^{-1} \end{aligned}$$

$\hookrightarrow v_1 = 2.765$

$$\begin{aligned} C_0(z) &= C_1(z) - v_1 B_1(z) \\ &= -0.76 + 2.765z^{-1} - 2.765(0.25 - z^{-1}) \\ &= -0.76 - 0.69125 \\ &= -1.45 \end{aligned}$$

$\rightarrow$

Lattice ladder:



Tutorial 41. S_0 :

$$\text{Here, } x[n] = v[n] - v[n-3]$$

$$= \{1, 1, 1, 0\}$$

$$\text{We know, } \vec{x}_N = \vec{W}_N \times \vec{x}_N$$

where, $N=4$

$$\text{So, } W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^1 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}_{4 \times 4} \quad x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = -j$$

$$W_4^2 = e^{-j\pi} = \cos\pi - j\sin\pi = -1$$

$$W_4^3 = e^{-j\frac{3\pi}{2}} = \cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2} = j$$

$$W_4^4 = -1$$

$$W_4^6 = e^{-j2\pi} = \cos 2\pi - j\sin 2\pi = 1$$

$$W_4^8 = e^{-j3\pi} = -1$$

$$W_4^9 = e^{-j\frac{9\pi}{2}} = -j$$

$$S_{0,} W_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -j \\ 1 & j & -1 & -j \end{bmatrix}$$

$$x_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$X_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -j \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1+1+1 \\ 1-j-1 \\ 1-1+j \\ 1+j-1 \end{bmatrix} = \begin{bmatrix} 3 \\ -j \\ 1 \\ j \end{bmatrix}$$

2. Circular Convolution:

$$x[n] = \{0, 1, 2, 3, 1\}, N=5$$

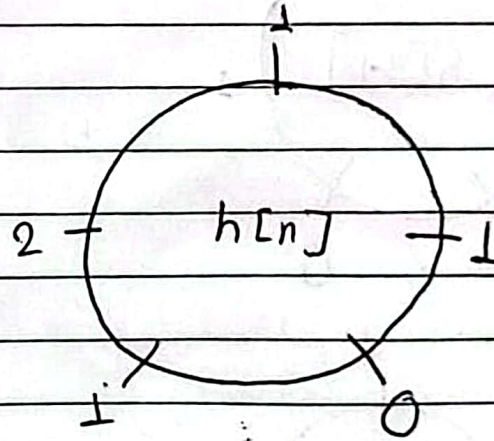
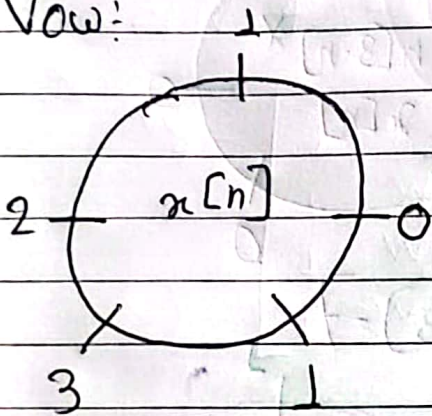
$$h[n] = \{1, 1, 2, 1\}, M=4$$

$$\text{o/p} \Rightarrow y[n] = \{y_0, y_1, y_2, y_3, y_4\}$$

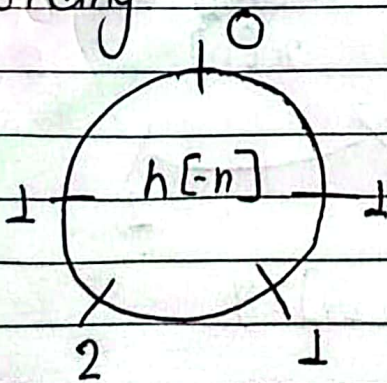
$$\text{So, } h[n] \text{ zero padding} \rightarrow \{1, 1, 2, 1, 0\}$$

Now:

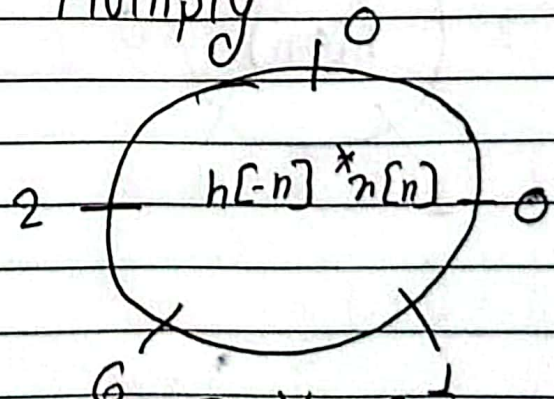
①



② Folding:

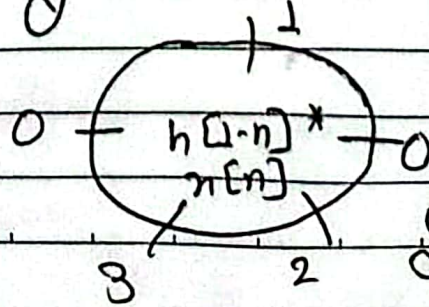
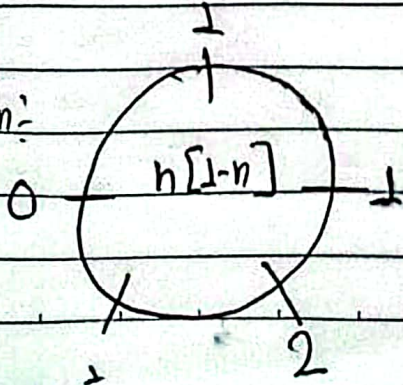


Multiply

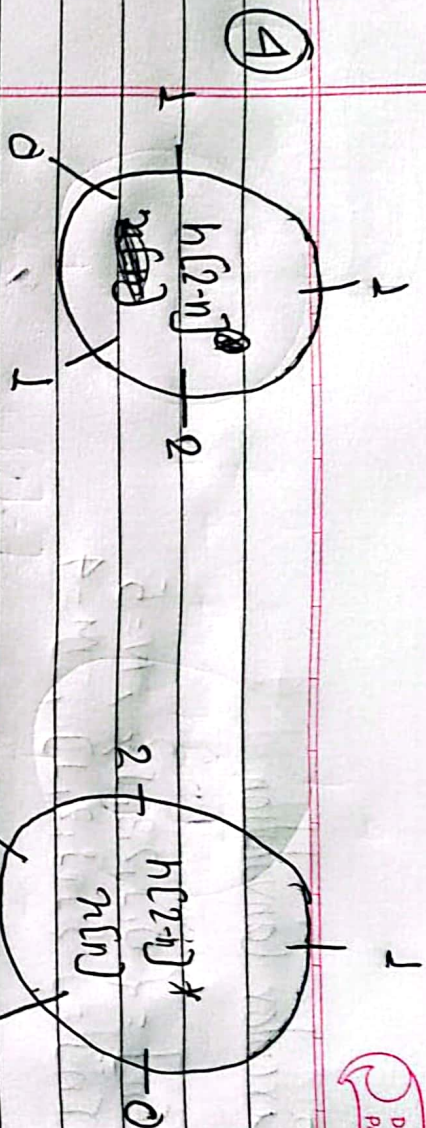


$$y[0] = y_0 = 9$$

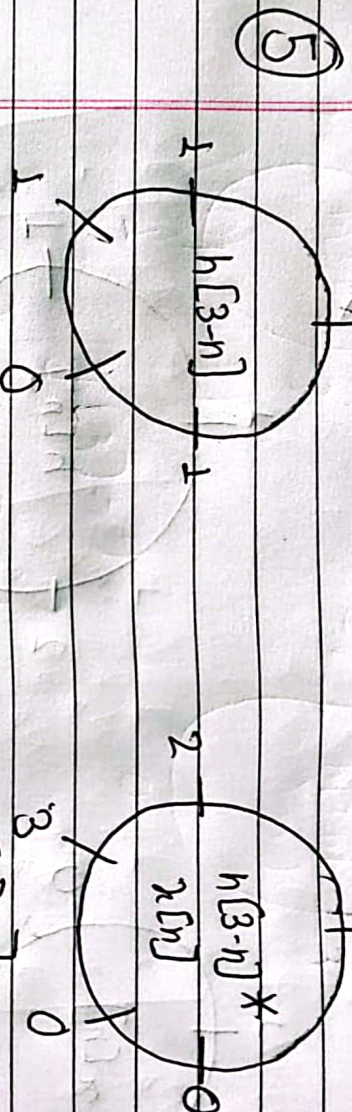
③ Again:



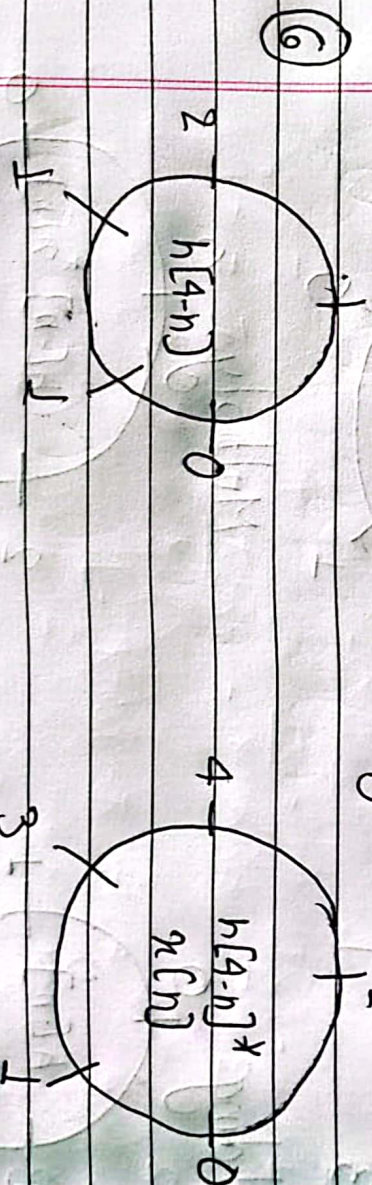
$$y[1] = 6$$



$$y[2] = 4$$



$$y[3] = 7$$



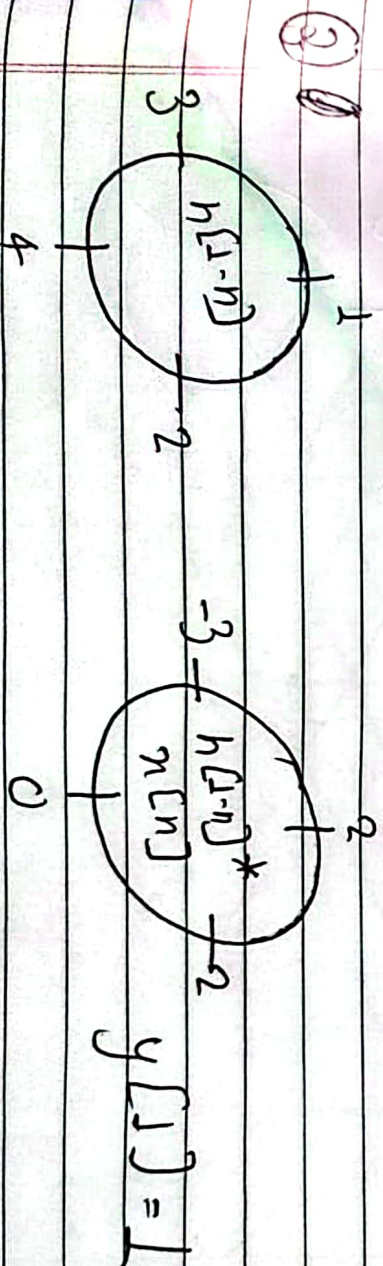
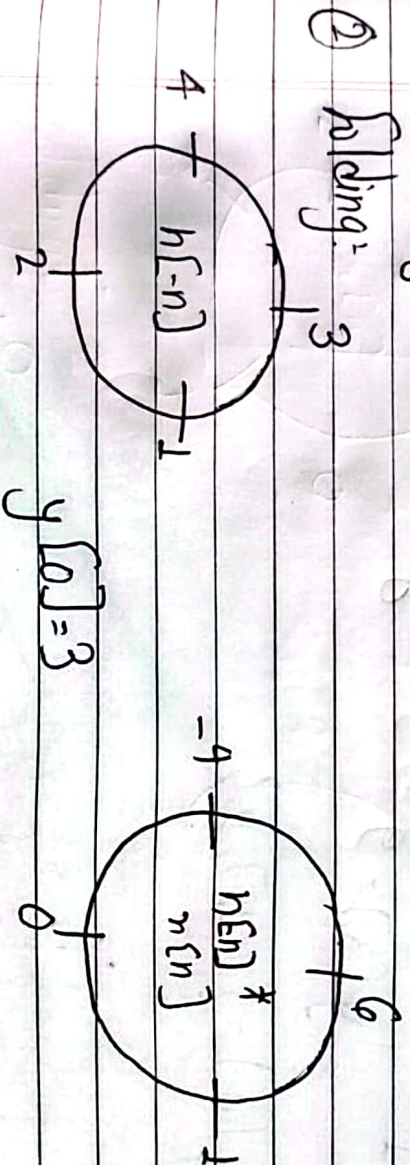
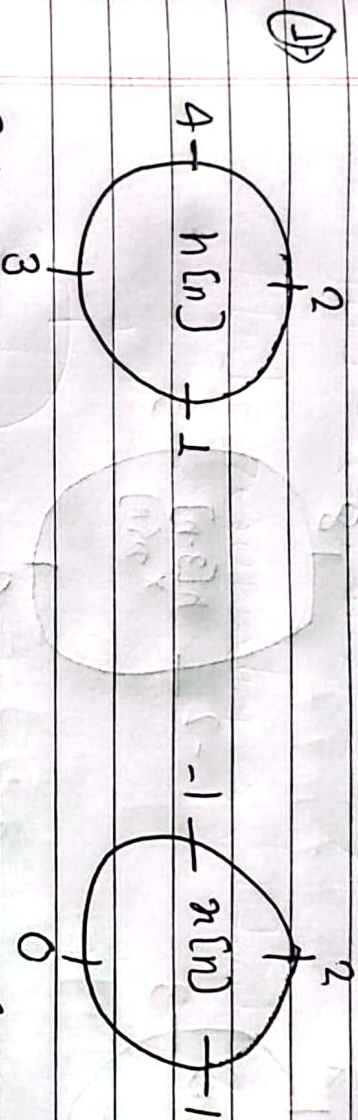
$$y[4] = 9$$

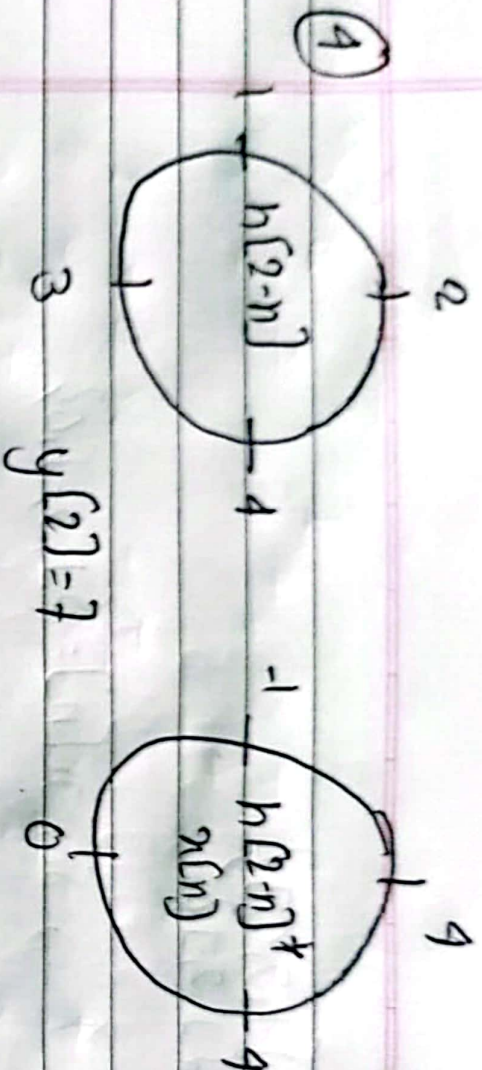
$$y[n] = [9, 6, 4, 7, 9] \neq$$

Qn-5) Solution:

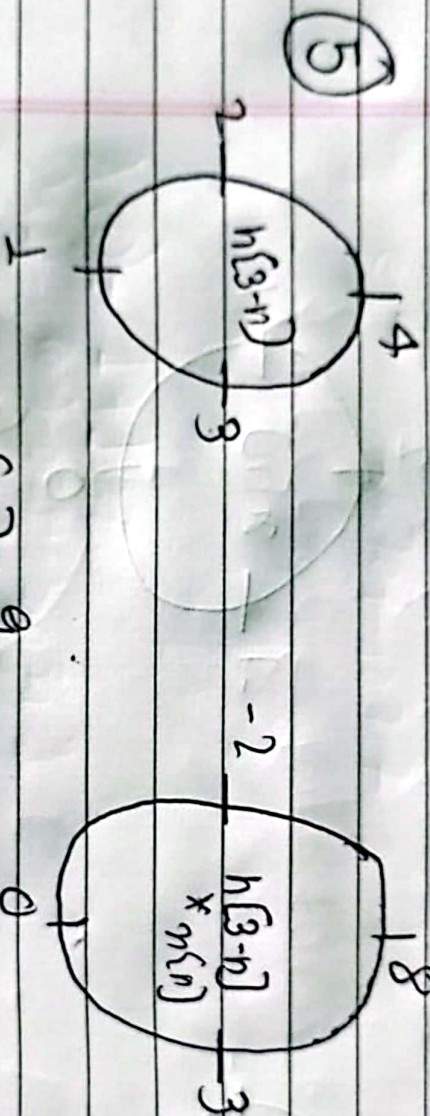
Here, $h[n] = \{1, 2, 4, 3\}$
 $x[n] = \{1, 2, 1\}$

length of $y[n] = 4$, so, $n[n] = \{1, 2, 1, 0\}$





$$y[2] = 7$$



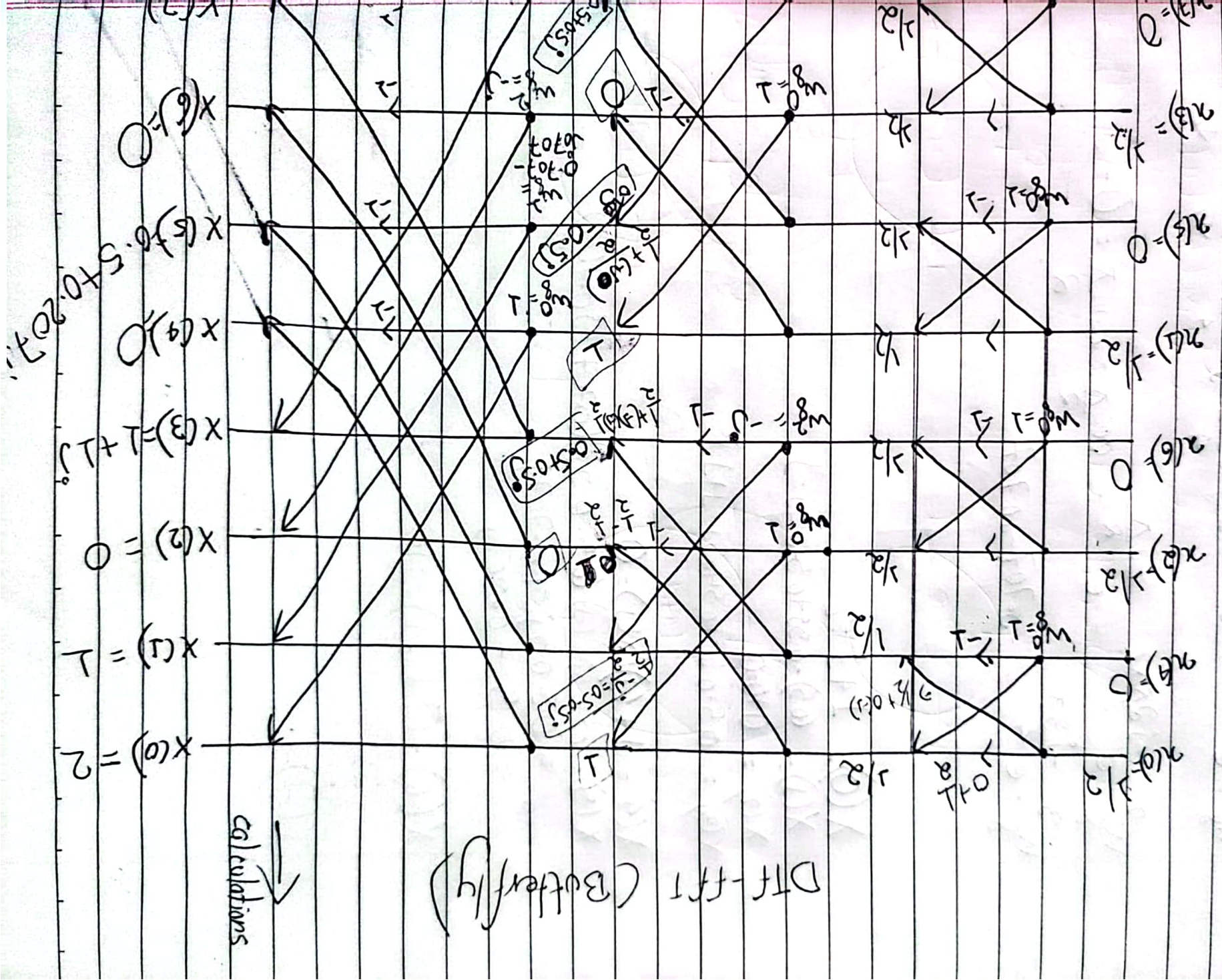
$$y[3] = 9$$

$$y[n] = \{3, 1, 7, 9\}$$

Q8-point DIT-FFT, $X[n] = \{ \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 1, 0, 0, 0 \}$

DIT-FFT (Butterfly)

calculations



$$W_8^0 = e^{-j \frac{2\pi \times 0}{8}} = 1$$

$$W_8^1 = e^{-j \frac{2\pi}{8}} = e^{-j \pi/4} = \cos \frac{\pi}{4} - j \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}} = 0.707 - j 0.707$$

$$W_8^2 = e^{-j \frac{2\pi \times 2}{8}} = e^{-j \frac{\pi}{2}} = \cos \frac{\pi}{2} - j \sin \frac{\pi}{2} = 0 - j$$

$$W_8^3 = e^{-j \frac{3\pi}{4}} = -0.707 - j 0.707$$

$$X(0) = 1 + 1 = 2$$

$$X(1) = (0.5) - 0.5j + 0.5 - 0.5j = 1$$

$$X(2) = 0 + 0 = 0$$

$$X(3) = 0.5 + 0.5j + 0.5 + 0.5j = 1 + 1j$$

$$X(4) = 1 + (-1)(1)(1) = 0$$

$$\begin{aligned} X(5) &= 0.5 - 0.5j + (-1)(0.5 - 0.5j)(0.707 - j 0.707) \\ &= 0.5 - 0.5j + 0.707j \\ &= 0.5 + 0.207j \end{aligned}$$

$$\begin{aligned} X(6) &= 0 + (-1)(-1)0 \\ &= 0 \end{aligned}$$

$$X(7) =$$

Tutorial-5Chap7 - IIR filter design.Impulse Invariance Method:

Transfer function in analog domain as:-

Derivation: We have impulse response of analog filter

$$H_a(s) = \sum_{k=1}^N \frac{C_k}{s - P_k} \dots (1)$$

Step1: We take Inverse Laplace transform of $H_a(s)$

$$\mathcal{L}^{-1}(H_a(s)) = \mathcal{L}^{-1}\left(\sum_{k=1}^N \frac{C_k}{s - P_k}\right)$$

$$h(t) = \mathcal{L}^{-1}\left(\frac{C_1}{s - P_1} + \frac{C_2}{s - P_2} + \dots + \frac{C_N}{s - P_N}\right)$$

$$= \mathcal{L}^{-1}\left(\frac{C_1}{s - P_1}\right) + \mathcal{L}^{-1}\left(\frac{C_2}{s - P_2}\right) + \dots + \mathcal{L}^{-1}\left(\frac{C_N}{s - P_N}\right)$$

We know, $\mathcal{L}^{-1}\left(\frac{1}{s - a}\right) = e^{at} u(t)$

$$\text{So, } h(t) = C_1 e^{P_1 t} u(t) + C_2 e^{P_2 t} u(t) + \dots + C_N e^{P_N t} u(t)$$

$$h(t) = \sum_{k=1}^N C_k e^{P_k t} u(t) \dots (2)$$

Step2: Sampling of $h(t)$

$$\text{put } t = nT_s$$

$$h(nT_s) = \sum_{k=1}^N C_k e^{P_k nT_s} u(nT_s) \dots (3)$$

Step 3: Taking Z Transform of ③

$$H(z) = \sum_{n=-\infty}^{\infty} h(nT_s) z^{-n}$$

We know,

$$U_n = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

$$\text{So, } H(z) = \sum_{n=0}^{\infty} \left[\sum_{k=1}^N C_k e^{P_k n T_s} \right] z^{-n}$$

$$= \sum_{k=1}^N C_k \left[\sum_{n=0}^{\infty} e^{P_k n T_s} z^{-n} \right]$$

$$= \sum_{k=1}^N C_k \left[\left(\sum_{n=0}^{\infty} e^{P_k T_s} z^{-1} \right)^n \right]$$

$$= \sum_{k=1}^N C_k \sum_{n=0}^{\infty} a^n$$

We know, $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}$

$$= \sum_{k=1}^N C_k \left[\frac{1}{1 - e^{P_k T_s} z^{-1}} \right] \dots \text{④}$$

Compare ① & ④

poles of ① $s = P_k$

poles of ② $e^{P_k T_s} = z$

$$\boxed{s \rightarrow z = e^{s T_s}} \dots \text{⑤}$$

$$\text{If } s = \sigma + j\omega$$

$$z = r e^{j\omega T}$$

$$\text{Put in (5): } Z \Rightarrow r e^{j\omega T} e^{(\sigma + j\omega)T}$$

$$= r e^{j\omega T} \cdot e^{\sigma T} \cdot e^{j\omega T}$$

Compare real & imag parts

$$r = e^{\sigma T}, \quad \omega = -\omega T$$

(*) Bilinear Transformation:

Let system fn of analog filter: $H(s) = \frac{b}{s+a}$... (1)

$$H(s) \Rightarrow \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$sY(s) + aY(s) = bX(s)$$

take inverse laplace:

$$\frac{dY(t)}{dt} + aY(t) = bX(t) \quad \dots (2)$$

Eqⁿ is integrated bet^m $nT-T$ & nT

$$\int_{nT-T}^{nT} \frac{dY(t)}{dt} dt + a \int_{nT-T}^{nT} Y(t) dt = b \int_{nT-T}^{nT} X(t) dt$$

We know, trapezoidal rule:

$$\int_{nT-T}^{nT} a(t) dt = \frac{T}{2} [a(nT) + a(nT-T)]$$

Apply on the eqⁿ:

$$Y(nT) - Y(nT-T) + \frac{aT}{2} Y(nT) + \frac{aT}{2} Y(nT-T) = \frac{bT}{2} Y(nT) + \frac{bT}{2} Y(nT-T) \quad \dots (4)$$

Taking Z transform:-

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + a} \quad \dots (5)$$

Comparing $H(z)$ with $H(s)$ we get

$$S = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + a \quad \dots (6)$$

put $z = re^{j\omega}$, $S = \sigma + j\omega$

$$S = \frac{2}{T} \frac{(re^{j\omega} - 1)}{re^{j\omega} + 1}$$

$$= \frac{2}{T} \left(\frac{r^2 - 1}{1 + r^2 + 2r \cos \omega} + \frac{j 2r \sin \omega}{1 + r^2 + 2r \cos \omega} \right) \quad \dots (7)$$

Consequently:

$$\sigma = \frac{2}{T} \frac{r^2 - 1}{1 + r^2 + 2r \cos \omega} \quad \dots (8)$$

$$\Omega = \frac{2}{T} \frac{2r \sin \omega}{1 + r^2 + 2r \cos \omega}$$

put $r=1, \sigma=0$ in Ω

$$\Omega = 2 \sin \omega$$

$$T \cos \omega$$

$$\epsilon_0 = 2 \tan \omega$$

$$\omega = \tan^{-1} \left(\frac{2}{T} \right) \cdot 2 \tan^{-1} \frac{\Omega T}{2}$$

2.

$$a) H(s) = \frac{s+0.2}{(s+0.2)^2 + 9}, T_s = 0.1$$

Here, analog filter has pole at $P_{1,2} = -0.2 \pm 3j$

By partial fraction:

$$H(s) = \frac{s+0.2}{(s+0.2)^2 + 3^2} = \frac{A}{s+0.2+3j} + \frac{A^*}{s+0.2-3j} \quad \text{--- (1)}$$

$$s+0.2 = A(s+0.2-3j) + A^*(s+0.2+3j)$$

Put $s = -0.2+3j$

$$-0.2+3j+0.2 = A^*(-0.2+3j+3j+0.2)$$

$$3j = A^* 6j$$

$$A^* = \frac{1}{2}$$

Put $s = -0.2-3j$

$$-0.2-3j+0.2 = A(-6j)$$

$$\frac{-3j}{-6j} = A$$

$$A = 1/2$$

So (1) becomes:

$$H(s) = \frac{1/2}{s+0.2+3j} + \frac{1/2}{s+0.2-3j}$$

(2) Take inverse laplace:

$$\mathcal{L}^{-1}\{H(s)\} = \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s+0.2+3j}\right) + \frac{1}{2} \mathcal{L}^{-1}\left(\frac{1}{s+0.2-3j}\right)$$

$$\begin{aligned}
 h(t) &= \frac{1}{2} \mathcal{L}^{-1} \left(\frac{1}{s - (-0.2 - 3j)} \right) + \frac{1}{2} \mathcal{L}^{-1} \left(\frac{1}{s - (-0.2 + 3j)} \right) \\
 &= \frac{1}{2} e^{(-0.2 - 3j)t} u(t) + \frac{1}{2} e^{(-0.2 + 3j)t} u(t) \\
 &= \frac{1}{2} u(t) (e^{(-0.2 - 3j)t} + e^{(-0.2 + 3j)t})
 \end{aligned}$$

Step 2: Sampling: $t = nT_s$

$$h(nT_s) = \frac{1}{2} u(nT_s) (e^{(-0.2 - 3j)nT_s} + e^{(-0.2 + 3j)nT_s})$$

Step 3: Z transform:

$$\begin{aligned}
 H(z) &= \sum_{n=0}^{\infty} \left[\frac{e^{(-0.2 - 3j)nT_s} + e^{(-0.2 + 3j)nT_s}}{2} \right] z^{-n} \\
 &= \frac{1}{2} \sum_{n=0}^{\infty} (e^{(-0.2 - 3j)T_s} z^{-1})^n + \frac{1}{2} \sum_{n=0}^{\infty} (e^{(-0.2 + 3j)T_s} z^{-1})^n
 \end{aligned}$$

$$\begin{aligned}
 H(z) &= \frac{1}{2} \left[\frac{1}{1 - e^{(-0.2 - 3j)T_s} z^{-1}} + \frac{1}{1 - e^{(-0.2 + 3j)T_s} z^{-1}} \right] \\
 &= \frac{1}{2} \left[\frac{1 - e^{(-0.2 + 3j)T_s} z^{-1}}{(1 - e^{(-0.2 - 3j)T_s} z^{-1})(1 - e^{(-0.2 + 3j)T_s} z^{-1})} + \frac{1 - e^{(-0.2 - 3j)T_s} z^{-1}}{(1 - e^{(-0.2 - 3j)T_s} z^{-1})(1 - e^{(-0.2 + 3j)T_s} z^{-1})} \right] \\
 &= \frac{z^{-1}}{2} \left[\frac{-e^{(-0.2 + 3j)T_s} - e^{(-0.2 - 3j)T_s}}{(1 - e^{(-0.2 - 3j)T_s} z^{-1})(1 - e^{(-0.2 + 3j)T_s} z^{-1})} \right]
 \end{aligned}$$

We have, $T_s = 0.1$

$$H(z) = \frac{z^{-1}}{2} \left[\frac{-e^{-(0.2+3j)0.1} - e^{-(0.2+3j)0.1}}{(1 - e^{-(0.2+3j)0.1} z^{-1})(1 - e^{-(0.2+3j)0.1} z^{-1})} \right]$$

$$= \frac{z^{-1}}{2} \left[-e^{-0.02} \cdot (e)^{-0.3j} - e^{-0.02} \cdot e^{-0.3j} \right]$$

$$s^2 + 2 \times 0.25s + 0.25 - 0.25 + 2$$

$$(s + 0.25)^2 - 1.9375$$

2.
① $H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$

$$\Rightarrow \frac{A}{s+0.5} + \frac{Bs+C}{s^2+0.5s+2}$$

$$1 = A(s^2+0.5s+2) + (Bs+C)(s+0.5)$$

Put $s = -0.5$

$$1 = A(2) + C(0.5) \quad 1 = A(0.25 - 0.25 + 2)$$

$$1 = A(2)$$

$$A = 1/2$$

Put $s =$

Put $s = 0$

$$1 = \frac{1}{2}(2) + C(0.5)$$

$$0 = C$$

Put $s = 1$

$$1 = \frac{1}{2}(1+0.5+2) + B(1.5)$$

$$1 = 1.75 + B(1.5)$$

$$-0.75 = B$$

$$\therefore H(s) = \frac{1/2}{s+0.5} - \frac{1/2 s}{s^2+0.5s+2}$$

$$h(t) = \frac{1}{2} e^{-0.5t} u(t) - \frac{1}{2} \mathcal{L}^{-1} \left(\frac{s}{s^2+0.5s+2} \right)$$

$$= \frac{1}{2} e^{-0.5t} u(t) - \frac{1}{2}$$

3.

$$H(s) = \frac{s+0.1}{(s+0.1)^2 + 9}, \quad \omega_r = \frac{\pi}{4}$$

Solⁿ:

$$H(z) = H(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)} \quad \dots (1)$$

From the eqn, $\omega_c = 3 \text{ rad/sec}$

$$\omega_c = \frac{2}{T} \tan \frac{\omega}{2}$$

$$T = \frac{2}{\omega_c} \tan \frac{\omega_r}{2}$$

$$= \frac{2}{3} \tan \frac{\pi}{8}$$

$$= 0.11$$

From (1)

$$H(z) = H(s) \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \frac{s+0.1}{(s+0.1)^2 + 9} \bigg|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= \frac{\frac{2}{0.11} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1}{\left(\frac{2}{0.11} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.1 \right)^2 + 9}$$

$$= \frac{18 \cdot 18 (1 - z^{-1}) + 1 + z^{-1}}{(18 \cdot 18 (1 - z^{-1}) + 1 + z^{-1})^2 + 9} \times \frac{1}{1 + z^{-1}}$$

$$= \frac{18 \cdot 18 - 18 \cdot 18 z^{-1} + 1 + z^{-1}}{(18 \cdot 18 - 18 \cdot 18 z^{-1} + 1 + z^{-1})^2 + 9(1 + z^{-1})^2} \times \frac{1}{(1 + z^{-1})}$$

$$= \frac{(19 \cdot 19 - 17 \cdot 18 z^{-1}) (1 + z^{-1})}{(18 \cdot 18 - 17 \cdot 18 z^{-1})^2 + 9 + 18 z^{-1} + 9 z^{-2}}$$

$$= \frac{19 \cdot 19 - 17 \cdot 18 z^{-1} + 19 \cdot 19 z^{-1} - 17 \cdot 18 z^{-2}}{(19 \cdot 18)^2 - 659 \cdot 02 z^{-1} + 295 \cdot 15 z^{-2} + 9 + 18 z^{-1} + 9 z^{-2}}$$

$$= \frac{19 \cdot 19 + 2 \cdot 72 z^{-1} - 17 \cdot 18 z^{-2}}{367 \cdot 87 - 641 \cdot 02 z^{-1} + 304 \cdot 15 z^{-2}}$$